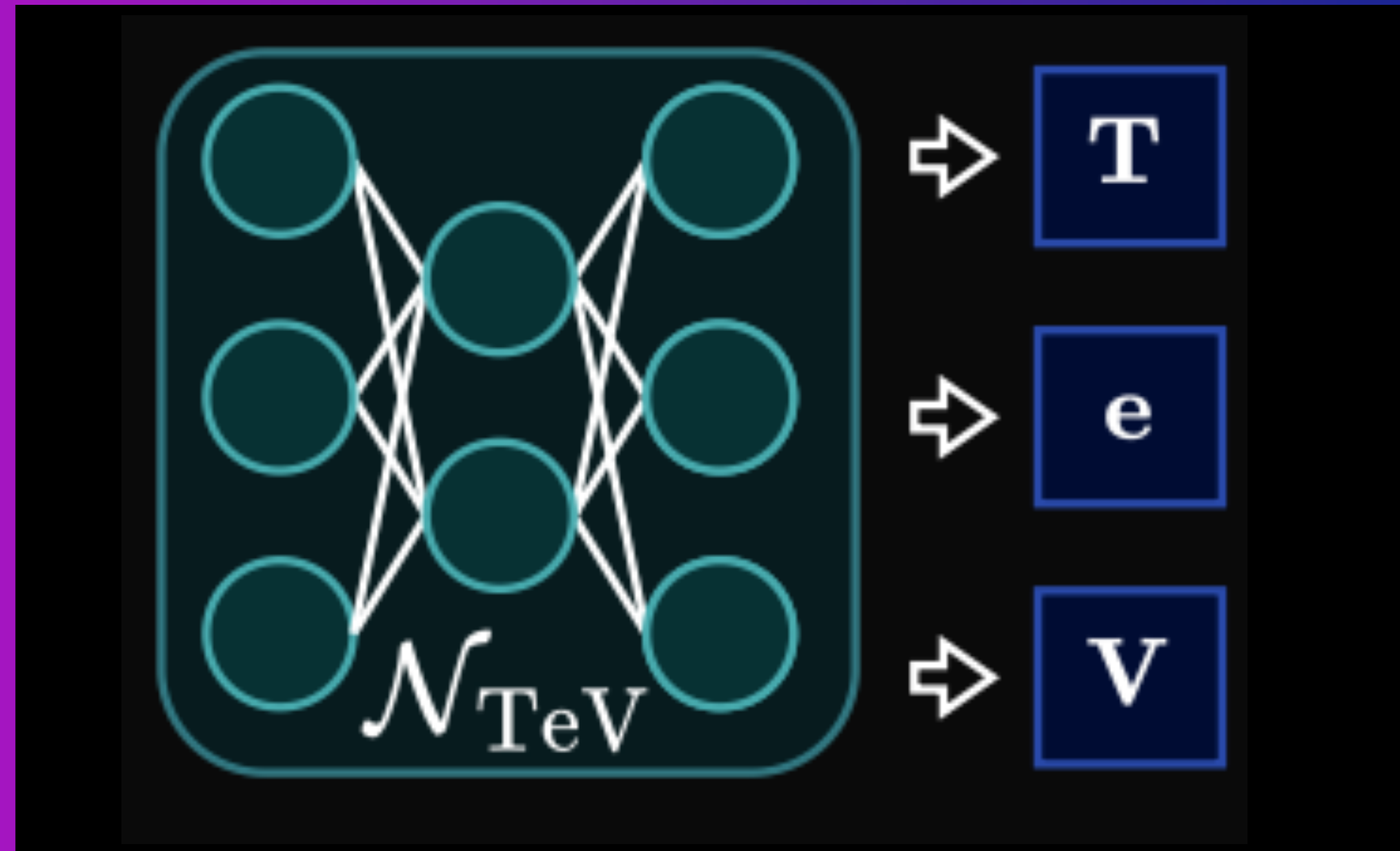


Physics-based Learning: Infrared Generation

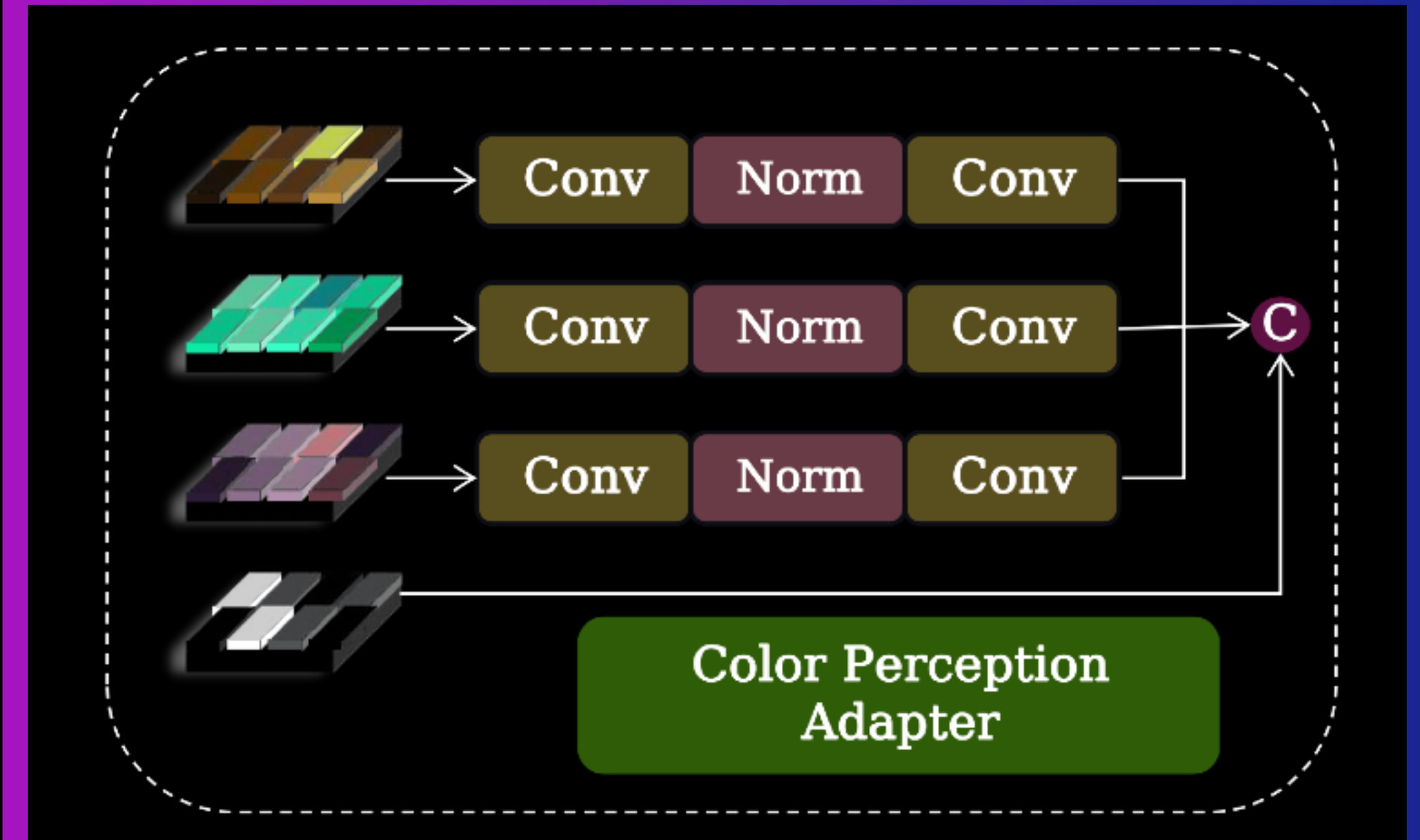
Diffusion Implementation



Physics-informed Diffusion for IR Generation

- Fangyuan Mao et. al. (July 2024).
- Introduce TeV Decomposition Network.
- Distinct architectural component - deploy in other settings.

Transformer Implementation

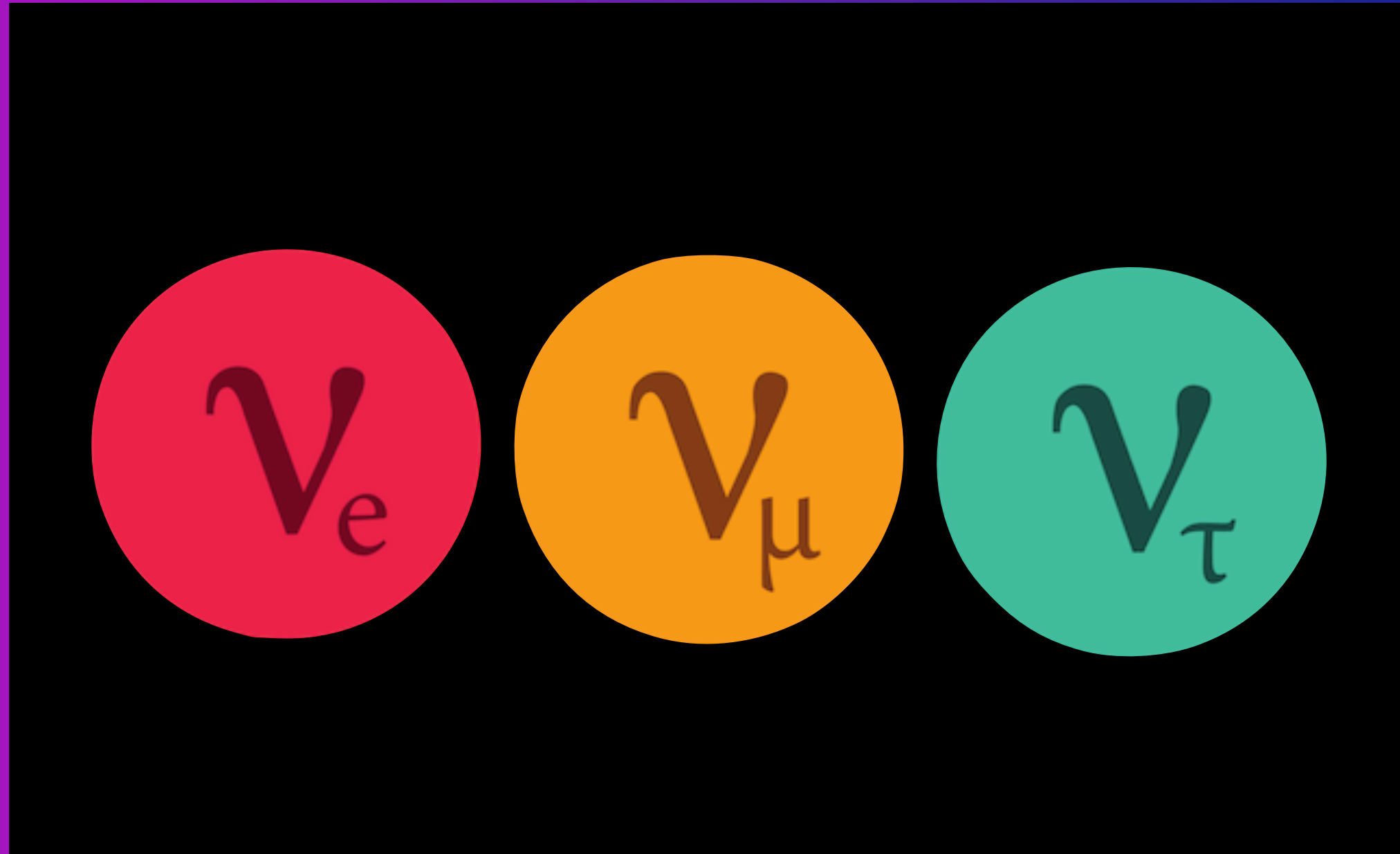


Physics-based Preprocessing

- Improve performance via image preprocessing.
- Deploy Enhanced Perception Attention Module.
- Transformer deals with global rendering.

Tau Neutrino Detection via GNNs

Neutrino Classification



GNN Classifier

- Solidify understanding of electron decay and the three-flavour paradigm
- Trained NuGraph2
- Implemented architectural modifications to improve performance.

